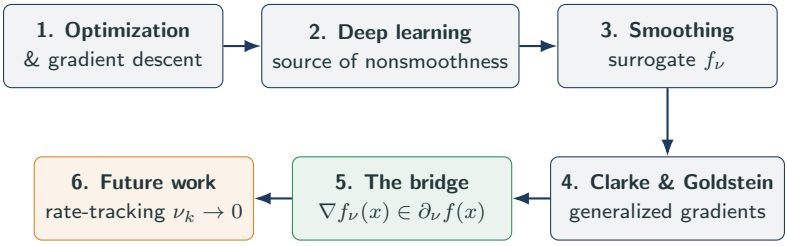
Randomized Smoothing for Nonsmooth Nonconvex Optimization

From gradient descent to Goldstein–Clarke stationarity,
and a proposal for rate-tracking ν

Jack Simon

2026

Outline



No prior background is assumed; all notation is defined before use.

Central question: *for a nonsmooth function, what is the appropriate notion of approximate stationarity?*

Optimization from zero

What is optimization?

- **Decision variables**: a vector $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$.
- **Objective function** $f : \mathbb{R}^n \to \mathbb{R}$ measures cost, error, or loss.
- Problem: find x attaining the smallest objective value

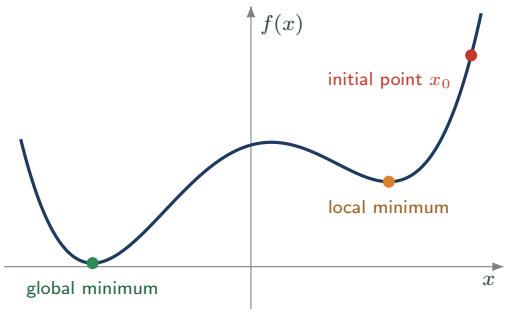
$$\min_{x \in \mathbb{R}^n} f(x).$$

variables x	**objective** $f(x)$
slope & intercept of a line	total distance to the data points
route choices	travel time
weights of a neural network	prediction error on the training data

Scale and access

In deep learning n may exceed 10^9. The function f is accessible only through local queries: values and (estimated) derivatives at individual points.

Geometric view: the graph of the objective



- Low values of f correspond to good decisions.
- **Global minimum**: lowest point anywhere.
  Local minimum: lowest point *nearby*.
- Algorithms observe only *local* information: the value $f(x)$ and the slope at the current point.

Derivatives and gradients

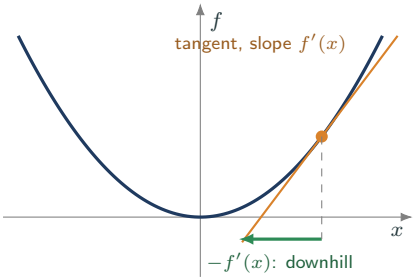
- $n = 1$: the **derivative** $f'(x)$ is the slope of the tangent line. Positive slope $\Rightarrow$ downhill is to the *left*.
- General n: the **gradient** collects the partial derivatives,

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \ldots, \frac{\partial f}{\partial x_n}(x) \right).$$

- Near x, f looks like a plane:

$$f(x + h) \approx f(x) + \nabla f(x)^\top h,$$

so $-\nabla f(x)$ is the direction of steepest descent.



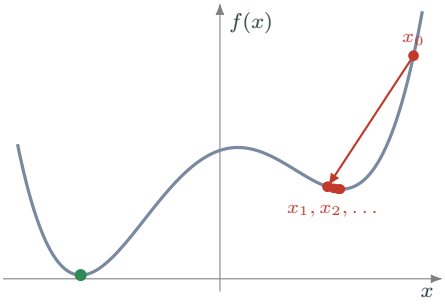
Gradient descent (GD)

Gradient descent

Pick a start x_0 and a step size $\gamma > 0$. Repeat

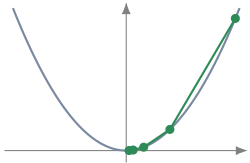
$$x_{k+1} \;=\; x_k \;-\; \gamma \nabla f(x_k).$$

- Each iteration moves in the direction of steepest local decrease.
- γ is the **step size** (in deep learning: *learning rate*).
- Here GD lands in the *local* minimum. For non-bowl-shaped (**nonconvex**) f that is normal, so we will need a more modest idea of success.

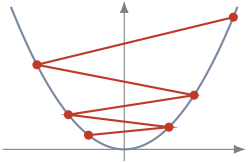


Step size and smoothness

$\gamma = 0.6$: monotone convergence



$\gamma = 1.8$: oscillation



- f is β-**smooth** if its gradient cannot change too fast:

$$\|\nabla f(x) - \nabla f(y)\| \leq \beta \|x - y\|.$$

  $\beta =$ "maximum curvature" (here $f = \frac{1}{2}x^2$, $\beta = 1$).

- **Descent lemma**: if $\gamma \leq 1/\beta$, every step makes progress,

$$f(x_{k+1}) \;\leq\; f(x_k) - \tfrac{\gamma}{2}\|\nabla f(x_k)\|^2.$$

- Large curvature β forces small step sizes.

Remark

Smoothing will trade *accuracy* against *curvature* β. That trade-off drives the future-work part.

Stationarity as a termination criterion

- A **stationary point** has flat ground: $\nabla f(x) = 0$. That includes minima, maxima and saddles.
- For nonconvex f the global minimum is out of reach in general, so the standard target is an ε-**stationary point**:

$$\|\nabla f(x)\| \leq \varepsilon.$$

- Classical guarantee: for β-smooth f, GD with $\gamma = 1/\beta$ finds one within

$$K \;\leq\; \frac{2\beta\,(f(x_0) - f^\star)}{\varepsilon^2} \quad \text{steps} \qquad \big(f^\star = \inf f\big).$$

(Sum the descent lemma over k; it telescopes.)

The question of this talk

What should "ε-stationary" mean when f is *nonsmooth*, so ∇f does not exist everywhere?

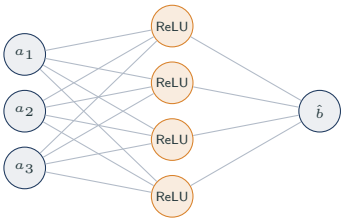
Deep learning and nonsmoothness

Deep learning as an optimization problem

- A neural network is a function $h_w(a)$: input a (image, text) $\mapsto$ prediction. The **weights** $w \in \mathbb{R}^n$ are the decision variables.
- Training data $(a_i, b_i)_{i=1}^N$. Loss

$$f(w) = \frac{1}{N} \sum_{i=1}^{N} \ell\big(h_w(a_i),\, b_i\big).$$

- Training a network $= \min_w f(w)$, solved by (variants of) gradient descent.



each line = one weight

Stochastic gradient descent (SGD)

- N is huge, so computing the exact ∇f at every step is too expensive.
- Instead use an inexpensive **random estimate** g_k that is right *on average*:

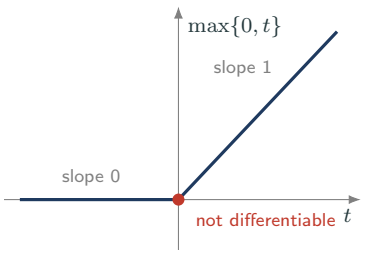
$$\mathbb{E}[g_k] = \nabla f(x_k) \qquad \text{(e.g. the gradient on a random mini-batch).}$$

- SGD: $x_{k+1} = x_k - \gamma_k\, g_k$.
- The noise averages out if the steps shrink suitably, e.g. $\sum_k \gamma_k = \infty$, $\sum_k \gamma_k^2 < \infty$ [Robbins–Monro 1951].

Key point

SGD only needs an *unbiased gradient estimate*. Later we build one for a *smoothed* function, using *function values only*.

Nonsmooth activations: ReLU

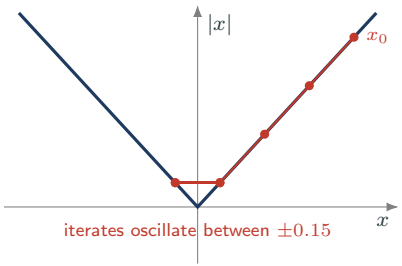


- Most networks use $\mathrm{ReLU}(t) = \max\{0, t\}$.
- A network composes thousands of ReLUs, so the loss $f(w)$ is **nonsmooth** (non-differentiable on a union of surfaces) and **nonconvex**.
- However, f is (locally) L-**Lipschitz**:

$$|f(x) - f(y)| \leq L \|x - y\|,$$

i.e. all difference quotients are bounded by L. This is our standing assumption.

Failure of smooth theory at nondifferentiable points



- $f(x) = |x|$: slope -1 or $+1$ everywhere, *undefined* at the minimizer 0.
- GD with a fixed step ($\gamma = 0.3$ here) oscillates about 0 and does not converge.
- Moreover, $|f'(x)| = 1$ at *every* $x \neq 0$. The stopping test $\|\nabla f(x)\| \leq \varepsilon$ never passes, even at $x = 10^{-9}$.
- No curvature bound either ($\beta = \infty$), so the rule "$\gamma \leq 1/\beta$" says nothing.

Zeroth-order (derivative-free) access

- **Black-box** f: a simulator, a physical experiment, non-differentiable code, hyper-parameter tuning, fine-tuning a model with forward passes only.
- We can only *evaluate* $f(x)$, possibly with noise. This is **zeroth-order** optimization.

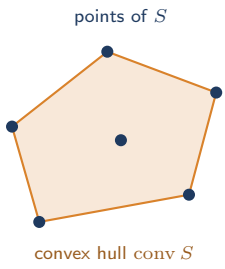
Requirements for the method

1. an SGD-type method that is valid for nonsmooth f,
2. that can run on function values only,
3. with a meaningful certificate of being "almost stationary".

Randomized smoothing

Notation

$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}$	length of a vector
$\mathbb{B} = \{x : \|x\| \le 1\}$	unit **ball** (solid)
$\mathbb{S}^{n-1} = \{x : \|x\| = 1\}$	unit **sphere** (just the surface)
$\xi \sim U, \quad \xi \sim V$	random point, uniform in $\mathbb{B}$ / on $\mathbb{S}^{n-1}$
$\mathbb{E}[\cdot]$	average over many random draws
L-Lipschitz	$\|f(x) - f(y)\| \le L\|x - y\|$
$\operatorname{conv} S$	all weighted averages of points of S
$\operatorname{dist}(0, S) = \min_{s \in S} \|s\|$	how close the set S gets to 0



The smoothed function f_ν

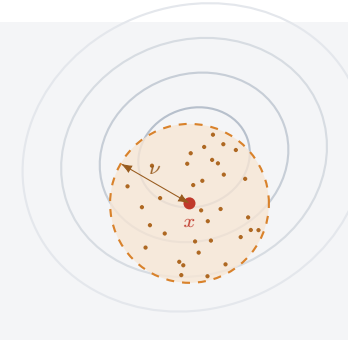
Ball smoothing (from the notes)

For a smoothing radius $\nu > 0$:

$$f_\nu(x) \;=\; \mathbb{E}_{\xi \sim U}\big[f(x + \nu\xi)\big],$$

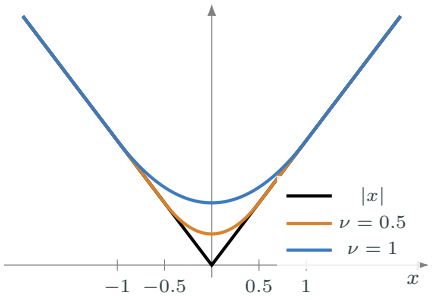
the *average* of f over the ball of radius ν around x.

- $f_\nu(x)$ is the local average of f over a ball of radius ν.
- Averaging removes points of nondifferentiability: f_ν is smooth even when f is not.
- Throughout: f is L-Lipschitz, nothing more.



contours of f; average over the dots

Example: smoothing $|x|$



In 1-D the ball is $[-1, 1]$, so $f_\nu(x) = \frac{1}{2\nu} \int_{x-\nu}^{x+\nu} |y|\, dy$:

$$f_\nu(x) = \begin{cases} \dfrac{x^2}{2\nu} + \dfrac{\nu}{2}, & |x| \le \nu, \\ |x|, & |x| > \nu, \end{cases}$$

$$f_\nu'(x) = \operatorname{clip}\big(x/\nu,\, -1,\, 1\big).$$

- The V becomes a U (the Huber function, lifted by $\nu/2$).
- The slope is now *small near the minimizer*, so stopping tests work again.

Properties of f_ν: accuracy versus curvature

Lemma (properties of f_ν; f is L-Lipschitz)

1. **Close to f:** $|f_\nu(x) - f(x)| \le L\nu$ for all x.
2. **Still Lipschitz:** f_ν is L-Lipschitz.
3. **Smooth:** ∇f_ν exists everywhere and is β_ν-Lipschitz with $\beta_\nu = \dfrac{\sqrt{n}\,L}{\nu}$.

[Yousefian–Nedić–Shanbhag 2012; Marrinan et al., Lem. 2.4]

Proof of 1, in one line: $|f(x+\nu\xi) - f(x)| \le L\nu\|\xi\| \le L\nu$ for every $\xi \in \mathbb{B}$; now average over ξ.

ν small

accurate (bias $\le L\nu$)
but very curved: $\beta_\nu \uparrow$, so steps $\gamma \lesssim \nu/(\sqrt{n}L)$

ν large

gentle landscape, big steps
but the wrong function: bias up to $L\nu$

Estimating ∇f_ν, recipe 1: average nearby gradients

Gradient representation (notes, second bullet)

$$\nabla f_\nu(x) = \mathbb{E}_{\xi \sim U}\big[\nabla f(x + \nu\xi)\big].$$

Estimator: draw ξ uniform in the ball and return $g = \nabla f(x + \nu\xi)$. Then $\mathbb{E}[g] = \nabla f_\nu(x)$.

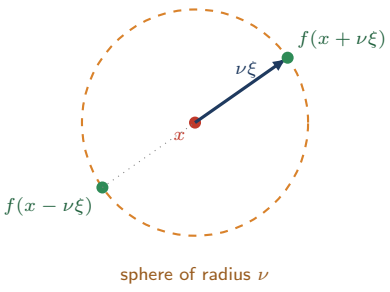
- Intuition: do not trust the slope *at* x (it may not exist); average the slopes around x.
- The notes assume $f \in C^1$, but the formula holds for any Lipschitz f. By **Rademacher's theorem** f is differentiable outside a set of zero volume, and $x + \nu\xi$ avoids that set with probability 1.
- This formula is the key to the Goldstein connection later.

Estimating ∇f_ν, recipe 2: function values only

Two-point representation ($\xi \sim V$ on the sphere)

$$\nabla f_\nu(x) = \frac{n}{2\nu}\, \mathbb{E}_{\xi\sim V}\Big[\big(f(x+\nu\xi) - f(x-\nu\xi)\big)\,\xi\Big].$$

- Evaluate f at two antipodal points: the quotient $(\cdot)/2\nu$ approximates the directional derivative along ξ; the factor n compensates for sampling one of n directions.
- **Why the sphere?** Divergence theorem: the gradient of a ball-average is a sphere-average of $f\times$(outward normal), $\nabla f_\nu(x) = \frac{n}{\nu}\mathbb{E}_V[f(x+\nu\xi)\xi]$; symmetrize $\xi \leftrightarrow -\xi$. (Backup slide.)
- **Bounded for every ν**: $\|g\| \le \frac{n}{2\nu}\cdot 2L\nu = nL$; finer: $\mathbb{E}\|g\|^2 \le c\,nL^2$ [Shamir 2017].



Smoothed SGD: the algorithm

Zeroth-order smoothed SGD

Input: black-box f, start x_0, radius ν, steps γ_k, budget K.

for $k = 0, 1, \ldots, K-1$:

1. draw ξ_k uniform on $\mathbb{S}^{n-1}$
2. $g_k = \frac{n}{2\nu}\big(f(x_k + \nu\xi_k) - f(x_k - \nu\xi_k)\big)\xi_k$
3. $x_{k+1} = x_k - \gamma_k\, g_k$

Output: x_R for a random index R (standard in nonconvex SGD theory), or the best iterate.

- Cost: 2 function evaluations per step; never touches ∇f.
- With recipe 1 instead: step 2 becomes $g_k = \nabla f(x_k + \nu\zeta_k)$ with ζ_k uniform in $\mathbb{B}$.
- Since $\mathbb{E}[g_k] = \nabla f_\nu(x_k)$ and f_ν is β_ν-smooth, **all standard SGD theory applies to** f_ν.

Smoothed SGD in Python (from the notes' task)

```python
import numpy as np

def smoothed_sgd(f, x0, nu, step, iters, seed=0):
    """SGD on f_nu(x) = E_{xi~U(ball)} f(x + nu*xi) via the two-point estimator.
    nu, step: numbers, or functions k -> value (rate-tracking)."""
    rng = np.random.default_rng(seed)
    nu_k   = nu   if callable(nu)   else (lambda k: nu)
    step_k = step if callable(step) else (lambda k: step)
    x = np.array(x0, dtype=float); n = x.size
    for k in range(iters):
        v  = nu_k(k)
        xi = rng.standard_normal(n)
        xi /= np.linalg.norm(xi)              # uniform on the unit sphere
        g  = n / (2*v) * (f(x + v*xi) - f(x - v*xi)) * xi
        x -= step_k(k) * g
    return x

f = lambda x: abs(x[0]) + 2*abs(x[1] - 1)     # nonsmooth, minimum at (0, 1)
x = smoothed_sgd(f, [3., -2.], nu=lambda k: (k+1)**-0.25,
                 step=lambda k: 0.5*(k+1)**-0.9, iters=20000)   # -> [0., 1.]
```

The full version (both estimators, mini-batches, logging) is in `smoothed_sgd.py`.

What the theory gives for a fixed ν

Standard SGD analysis on the β_ν-smooth f_ν, with $\mathbb{E}\|g\|^2 \le \sigma^2$ and $\Delta_\nu = f_\nu(x_0) - \inf f_\nu$:

$$\frac{1}{K}\sum_{k<K} \mathbb{E}\|\nabla f_\nu(x_k)\|^2 \;\le\; \frac{\Delta_\nu}{\gamma K} + \frac{\beta_\nu \gamma \sigma^2}{2} \quad \xrightarrow{\text{best } \gamma} \quad \sqrt{\frac{2\Delta_\nu \beta_\nu \sigma^2}{K}}.$$

Plug in $\beta_\nu = \sqrt{n}L/\nu$ and $\sigma^2 \le c\,nL^2$:

$$\mathbb{E}\|\nabla f_\nu(x_R)\| \le \varepsilon \quad \text{after} \quad K = O\!\left(\frac{n^{3/2}L^3\,\Delta_\nu}{\nu\,\varepsilon^4}\right) \text{ steps}$$

(the same order as Lin–Zheng–Jordan 2022: $O(n^{3/2}\delta^{-1}\varepsilon^{-4})$).

But...

We can make ∇f_ν small. f_ν is not the function we care about.
What does a small $\nabla f_\nu(x)$ tell us about f itself?

Clarke and Goldstein generalized gradients

Step 1: Lipschitz functions are differentiable almost everywhere

Rademacher's theorem

An L-Lipschitz $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at every point outside a set of *zero volume*. Write D_f for the set of points where ∇f exists.

- ReLU network: the nondifferentiable set is a union of surfaces of measure zero; a random point avoids it with probability 1.
- Wherever ∇f exists, $\|\nabla f\| \leq L$.
- So gradients exist *almost* everywhere. The difficulty is at and near the nondifferentiable set, where minimizers frequently lie (e.g. $|x|$).

The Clarke subdifferential

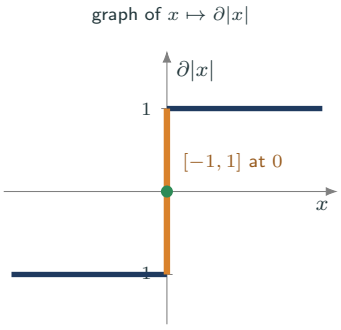
Definition [Clarke 1983]

$$\partial f(x) = \operatorname{conv}\left\{ \lim_{k\to\infty} \nabla f(x_k) \;:\; x_k \to x,\; x_k \in D_f \right\}.$$

Read: walk toward x through points where f is differentiable, collect all limiting gradients, and take the convex hull.

- f smooth at x: $\partial f(x) = \{\nabla f(x)\}$.
- $|x|$ at 0: slopes $\pm 1 \Rightarrow \partial f(0) = [-1, 1]$.
- ReLU at 0: $[0, 1]$. $\quad -|x|$ at 0: $[-1, 1]$.

Clarke stationary: $0 \in \partial f(x)$. This generalizes $\nabla f(x) = 0$, maxima included (as for $-|x|$ at 0).



Approximate Clarke stationarity is (too) hard

- Natural target: ε-**Clarke stationary**, $\operatorname{dist}\big(0, \partial f(x)\big) \le \varepsilon$.
- For $|x|$: $\partial f(x) = \{\operatorname{sign} x\}$ for every $x \neq 0$, so $\operatorname{dist}(0, \partial f(x)) = 1$. Being 10^{-9} away from the minimizer earns nothing.
- This is not specific to the example:

Negative result [Zhang–Lin–Jegelka–Sra–Jadbabaie, ICML 2020]

For Lipschitz (even directionally differentiable) f, no first-order method can guarantee an ε-Clarke-stationary point in a finite number of steps.

- Remedy: allow the certificate to use gradients from a small neighbourhood of x, not just from x itself.

The Goldstein δ-subdifferential ("δ-Clarke generalized gradient")

Definition [Goldstein 1977]

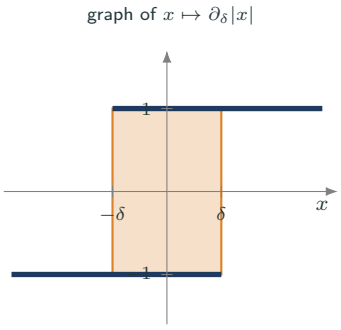
$$\partial_\delta f(x) = \operatorname{conv}\Big(\bigcup_{\|y-x\| \le \delta} \partial f(y) \Big).$$

x is (δ, ε)-**Goldstein stationary** if $\operatorname{dist}\big(0, \partial_\delta f(x)\big) \le \varepsilon$.

Read: the convex hull of all Clarke subgradients in the δ-ball around x. Small $\operatorname{dist}(0, \partial_\delta f(x))$ means some *convex combination of nearby subgradients* has small norm.

Example $|x|$: $\partial_\delta |\cdot|(x) = \begin{cases} [-1,1], & |x| \le \delta, \\ \{\operatorname{sign} x\}, & |x| > \delta. \end{cases}$

Every point within δ of 0 is $(\delta, 0)$-stationary.



Clarke vs. Goldstein, side by side

	Clarke $\partial f(x)$	**Goldstein** $\partial_\delta f(x)$
built from	limits of slopes at points $\to x$	slopes at *all* points within δ
the set is	nonempty, convex, compact, $\subseteq L\mathbb{B}$	nonempty, convex, compact, $\subseteq L\mathbb{B}$
if f is smooth	$\{\nabla f(x)\}$	$\subseteq \nabla f(x) + \beta\delta\,\mathbb{B}$ (if ∇f is β-Lipschitz)
stationarity	$0 \in \partial f(x)$	$\operatorname{dist}(0, \partial_\delta f(x)) \le \varepsilon$
reachable in finite time?	no [Zhang et al. 2020]	yes: $\tilde{O}(\delta^{-1}\varepsilon^{-3})$ first-order [Zhang et al.]; $O(n\delta^{-1}\varepsilon^{-3})$ zeroth-order [Kornowski–Shamir 2024]

How they are related

- **Containment:** $\partial f(x) \subseteq \partial_\delta f(x)$ (take $y = x$).
- **Monotone:** $\delta_1 \le \delta_2 \;\Rightarrow\; \partial_{\delta_1} f(x) \subseteq \partial_{\delta_2} f(x)$.
- **Limit:** $\partial_\delta f(x) \downarrow \partial f(x)$ as $\delta \downarrow 0$ (next slide).

Goldstein $\to$ Clarke as $\delta \to 0$

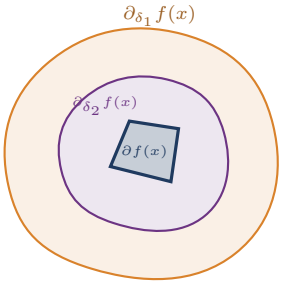
Theorem [Goldstein 1977]

$$\bigcap_{\delta>0} \partial_\delta f(x) = \partial f(x).$$

Proof. $\supseteq$ is containment. For $\subseteq$: Clarke's ∂f is upper semicontinuous, so for every $\eta > 0$ there is $\delta > 0$ with $\partial f(y) \subseteq \partial f(x) + \eta\mathbb{B}$ whenever $\|y - x\| \le \delta$. The right side is convex, so the convex hull stays inside: $\partial_\delta f(x) \subseteq \partial f(x) + \eta\mathbb{B}$. Let $\eta \to 0$. □

Robust version. If $x_k \to \bar{x}$, $\delta_k \to 0$ and $\operatorname{dist}(0, \partial_{\delta_k} f(x_k)) \to 0$, then $0 \in \partial f(\bar{x})$.
(Use $\partial_{\delta_k} f(x_k) \subseteq \partial_{\delta_k + \|x_k - \bar{x}\|} f(\bar{x})$.)

Smooth check. If ∇f is β-Lipschitz, (δ, ε)-Goldstein gives $\|\nabla f(x)\| \le \varepsilon + \beta\delta$. Take $\delta = \varepsilon/\beta$ to recover ordinary ε-stationarity.



Goldstein is a relaxation of Clarke at scale δ.

The bridge: smoothing meets Goldstein

The bridge theorem

Theorem [Lin–Zheng–Jordan 2022; Marrinan–Shanbhag–Yousefian, Prop. 2.5]

If f is L-Lipschitz and $\nu > 0$, then for every x

$$\nabla f_\nu(x) \;\in\; \partial_\nu f(x).$$

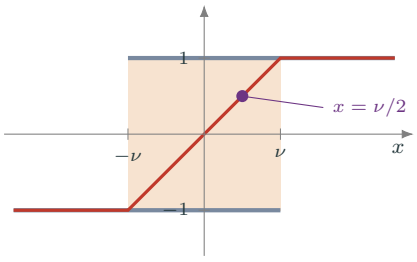
Proof in three lines.

1. $\nabla f_\nu(x) = \mathbb{E}_{\xi\sim U}[\nabla f(x+\nu\xi)]$ (recipe 1, valid a.e. by Rademacher).
2. Each sample $\nabla f(x+\nu\xi) \in \partial f(x+\nu\xi) \subseteq \bigcup_{\|y-x\|\le\nu} \partial f(y)$, since $\|\nu\xi\| \le \nu$.
3. An average of points of a set lies in its convex hull, so $\mathbb{E}[\,\cdot\,] \in \partial_\nu f(x)$. □

Answer to the question in the notes

$\|\nabla f_\nu(x)\| \le \varepsilon \implies \operatorname{dist}\big(0, \partial_\nu f(x)\big) \le \varepsilon$: an ε-stationary point of f_ν is (ν,ε)-Goldstein stationary for f. **The smoothing radius is the Goldstein radius.**

Example: $f(x) = |x|$



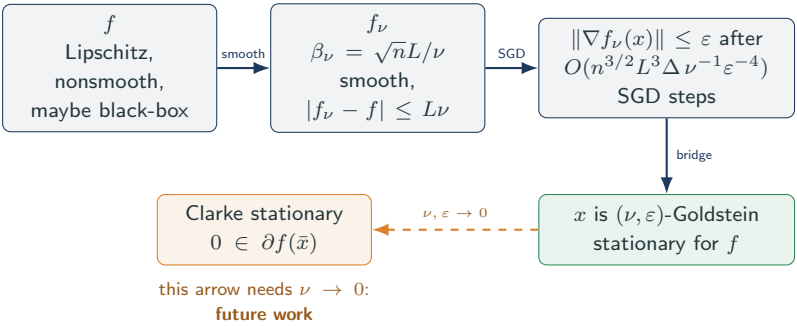
- $|x| \le \nu$: $f'_\nu(x) = x/\nu \in [-1, 1] = \partial_\nu f(x)$. ✓
- $|x| > \nu$: $f'_\nu(x) = \operatorname{sign} x$, and $\partial_\nu f(x) = \{\operatorname{sign} x\}$. ✓

The converse fails

$x = \nu/2$ is $(\nu, 0)$-Goldstein stationary (since $0 \in [-1, 1]$), yet $f'_\nu(\nu/2) = \frac{1}{2}$.

Small $\nabla f_\nu \Rightarrow$ Goldstein stationary, *not* the reverse. The smoothing certificate is sufficient, not necessary.

Putting it together



- Known complexity for (δ,ε)-Goldstein points from function values: $O(n^{3/2}\delta^{-1}\varepsilon^{-4})$ [Lin et al. 2022], improved to the optimal $O(n\delta^{-1}\varepsilon^{-3})$ [Kornowski–Shamir 2024]; constrained, variance-reduced and quasi-Newton versions in [Marrinan et al.].
- In all of them ν is fixed in advance.

Future work: rate-tracking ν

Limitations of a fixed smoothing radius

- The guarantee holds at a single scale ν: **Goldstein, not Clarke**. Approaching Clarke stationarity requires rerunning with smaller ν.
- ν, the step size and the horizon K are all chosen *before* running (the best γ scales like $\sqrt{\nu/K}$).
- The bias $|f_\nu - f| \leq L\nu$ never goes away.
- Small ν means huge curvature ($\beta_\nu = \sqrt{n}L/\nu$), so tiny steps from iteration 1.
- With **noisy** function values $f(x) + e$, $\operatorname{Var} e = \sigma_e^2$, the estimator variance grows like $n^2\sigma_e^2/\nu^2$.

Summary

A fixed ν settles the bias / curvature / variance trade-off *once, at one scale, in advance*.

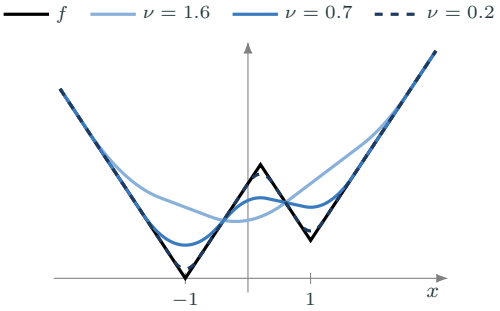
Proposal: iteration-dependent smoothing $\nu_k \to 0$

Rate-tracking smoothed SGD

$$x_{k+1} = x_k - \gamma_k \, g_{\nu_k}(x_k), \qquad \nu_k \downarrow 0,$$

with γ_k (and batch size m_k) shrinking at rates coupled to ν_k.

- Early iterations: large ν_k, larger steps, robustness to nonsmoothness and noise.
- Late iterations: small bias, and the Goldstein radius $\nu_k \to 0$.



$f(x) = \min\{|x+1|,\ 0.4 + |x-1|\}$: at $\nu = 1.6$ the two valleys merge

Precedents. Kiefer–Wolfowitz 1952 (finite differences, $c_k \to 0$, $\sum a_k^2/c_k^2 < \infty$, smooth f); Chen 2012 (deterministic smoothing, μ_k updates $\to$ Clarke points); Chen–Kelley–Xu–Zhang 2018 (direct search with coupled stencil / smoothing / sample-size rates). **Gap:** stochastic zeroth-order *ball* smoothing for general Lipschitz nonconvex f with explicit (γ_k, ν_k, m_k) rates.

Descent analysis with varying ν_k

Two facts. (a) one SGD step on the *current* surrogate f_{ν_k} (descent lemma, $\beta_{\nu_k} = \sqrt{n}L/\nu_k$):

$$\mathbb{E}\big[f_{\nu_k}(x_{k+1}) \,\big|\, x_k\big] \le f_{\nu_k}(x_k) - \gamma_k \|\nabla f_{\nu_k}(x_k)\|^2 + \tfrac{\sqrt{n}L}{2\nu_k}\gamma_k^2\sigma_k^2;$$

(b) the surrogates vary continuously in ν: $|f_\nu(x) - f_\mu(x)| \le L|\nu - \mu|$.

Lyapunov function. With ν_k decreasing, $V_k := f_{\nu_k}(x_k) + L\nu_k - f^\star \ge 0$ absorbs the drift:

$$\boxed{\mathbb{E}[V_{k+1} \,|\, x_k] \;\le\; V_k - \gamma_k \|\nabla f_{\nu_k}(x_k)\|^2 + \frac{\sqrt{n}L}{2}\,\frac{\gamma_k^2\sigma_k^2}{\nu_k}}$$

Tracking condition: $\sum_k \gamma_k = \infty$ **and** $\sum_k \gamma_k^2\sigma_k^2/\nu_k < \infty$

- exact function values ($\sigma_k^2 \le n^2L^2$): $\sum_k \gamma_k^2/\nu_k < \infty$
- noisy values, batch m_k ($\sigma_k^2 \lesssim n^2L^2 + n^2\sigma_e^2/(m_k\nu_k^2)$): $\sum_k \gamma_k^2/(m_k\nu_k^3) < \infty$

Consequence: asymptotic Clarke stationarity (sketch)

Proposition (sketch; to be made rigorous)

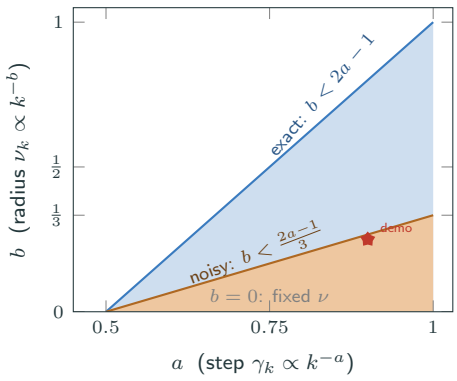
Let f be L-Lipschitz and bounded below, $\nu_k \downarrow 0$, the tracking condition hold, and the iterates stay bounded. Then almost surely:

1. $\sum_k \gamma_k \|\nabla f_{\nu_k}(x_k)\|^2 < \infty$ (Robbins–Siegmund on V_k), hence $\liminf_k \|\nabla f_{\nu_k}(x_k)\| = 0$;
2. since $\nabla f_{\nu_k}(x_k) \in \partial_{\nu_k} f(x_k)$ and $\nu_k \to 0$, the robust Goldstein$\to$Clarke lemma gives an accumulation point $\bar{x}$ with

$$0 \in \partial f(\bar{x}) \qquad \text{(Clarke stationary)}.$$

This is exactly the dashed arrow: Goldstein at radius ν_k along the run, Clarke in the limit, and no ν chosen in advance. Still to check: *every* accumulation point (not just one), and whether a similar result is already in the literature.

How fast may ν shrink?



Polynomial schedules $\gamma_k = \gamma_0(k+1)^{-a}$, $\nu_k = \nu_0(k+1)^{-b}$, $\frac{1}{2} < a \le 1$:

- exact values: $\sum \gamma_k^2/\nu_k < \infty \iff b < 2a - 1$
- noisy values, $m_k \equiv 1$: $b < (2a-1)/3$

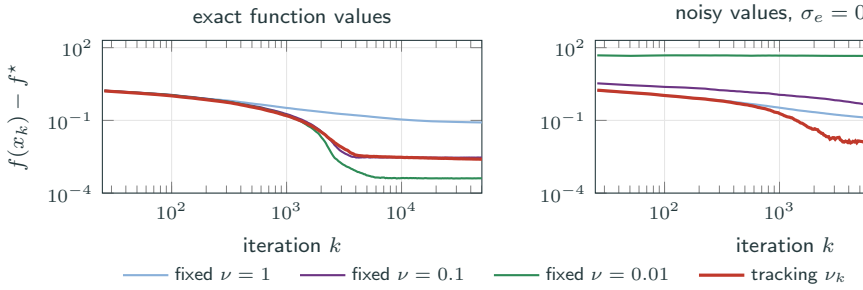
Rate heuristic (exact values). Sample the output $R \propto \gamma_k$ on $[K/2, K)$:

$$\mathbb{E}\|\nabla f_{\nu_R}(x_R)\|^2 = O\big(K^{-(1-a)}\big), \qquad \nu_R \le \nu_0(K/2)^{-b}.$$

Near the edge $a \approx \frac{1+b}{2}$ this is $K \approx 1/(\nu\varepsilon^4)$ up to logs: the same (ν, ε) order as fixed-ν SGD with optimally tuned ν, without fixing ν or K in advance.

Numerical illustration

$f(x) = \frac{1}{N}\|Ax - b\|_1$ ($n = 20$, $N = 100$, 20% outliers; $f^\star$ by LP). Same $\gamma_k = (k+1)^{-0.9}$ in all runs; tracking $\nu_k = (k+1)^{-1/4}$; median of 10 runs.



Exact: $\nu = 1$ stalls at a bias floor; small ν is free here ($\|g\| \le nL$), so $\nu = 0.01$ wins.

Noisy: $\nu = 0.01$ is variance-dominated, $\nu = 1$ bias-dominated; tracking leads at every checkpoint ($4.6{\cdot}10^{-3}$ vs. $6.0{\cdot}10^{-3}$ for $\nu = 0.1$ at $k = 5{\cdot}10^4$).

Open questions and research plan

1. **Rigor.** Make the proposition precise: is *every* accumulation point Clarke stationary? Can the bounded-iterates assumption be dropped? Do a careful literature check.
2. **Rates.** Non-asymptotic (ν_k, ε_k) guarantees; best (a, b) and batch growth m_k; dimension dependence. Can tracking reach the optimal $O(n\delta^{-1}\varepsilon^{-3})$ of Kornowski–Shamir instead of plain SGD's ε^{-4}?
3. **Adaptive tracking.** Shrink ν when a mini-batch estimate of $\|\nabla f_\nu\|$ drops below $c\,\nu$ (in the spirit of Chen 2012). This needs a statistically reliable test.
4. **Beyond SGD.** Variance reduction and quasi-Newton (VRG-ZO, VRSQN-ZO of Marrinan et al.) with $\nu_k \to 0$: what happens to Hessian approximations as $\beta_{\nu_k} \to \infty$?
5. **Experiments.** ReLU networks, constrained problems, genuinely noisy simulators.

Summary

- Nonsmoothness (e.g. ReLU) and zeroth-order access invalidate the standard $\|\nabla f\| \leq \varepsilon$ criterion.
- **Smoothing** $f_\nu = \mathbb{E} f(x + \nu\xi)$: smooth ($\beta_\nu = \sqrt{n}L/\nu$), close ($L\nu$), and estimable from two function values.
- **Clarke** ∂f: limiting slopes, filled in. **Goldstein** $\partial_\delta f$: slopes from a δ-ball, filled in. $\partial f \subseteq \partial_\delta f$, and $\partial_\delta f \to \partial f$ as $\delta \to 0$.
- **Bridge**: $\nabla f_\nu(x) \in \partial_\nu f(x)$, so a small smoothed gradient means (ν, ε)-Goldstein stationarity.
- **Future**: let $\nu_k \to 0$ at a rate the step size tracks ($\sum \gamma_k^2/\nu_k < \infty$) to upgrade Goldstein to Clarke without fixing ν in advance.

Questions

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Backup: where the two-point formula comes from

Write $f_\nu(x) = \frac{1}{\mathrm{vol}(\nu\mathbb{B})} \int_{\nu\mathbb{B}} f(x+u)\,du$ and differentiate under the integral:

$$\nabla f_\nu(x) = \frac{1}{\mathrm{vol}(\nu\mathbb{B})} \int_{\nu\mathbb{B}} \nabla f(x+u)\,du \overset{\text{divergence thm}}{=} \frac{1}{\mathrm{vol}(\nu\mathbb{B})} \int_{\nu\mathbb{S}^{n-1}} f(x+u)\,\frac{u}{\nu}\,dS(u).$$

Since $\mathrm{area}(\nu\mathbb{S}^{n-1})/\mathrm{vol}(\nu\mathbb{B}) = n/\nu$, this is

$$\nabla f_\nu(x) = \frac{n}{\nu}\,\mathbb{E}_{\xi\sim V}\big[f(x+\nu\xi)\,\xi\big].$$

Because $\mathbb{E}_V[\xi] = 0$ and $\xi \overset{d}{=} -\xi$,

$$\frac{n}{\nu}\mathbb{E}_V[f(x+\nu\xi)\xi] = \frac{n}{2\nu}\mathbb{E}_V\big[(f(x+\nu\xi) - f(x-\nu\xi))\,\xi\big].$$

The symmetric form subtracts the "baseline" $f(x)$ implicitly, which is why its size is $O(nL)$ rather than $O(n|f(x)|/\nu)$.